MANAGEMENT OF AI PRODUCTS - MBA ZG583

EXPERIENTIAL LEARNING ASSIGNMENT TEAM 002

CAREPULSE - Diabetes Safety & Doctor Consultation Companion

Executive Summary:

Business Problem:

Type 2 diabetes mellitus is a major chronic health challenge in India. A nationally representative survey of adults aged 45 years and older reported a diabetes prevalence of 19.8%, with ~50.4 million individuals in this age group. [1] Many individuals with type 2 diabetes also have other comorbidities such as hypertension, dyslipidemia, cardiovascular disease, chronic kidney disease, neuropathy, and other diabetes-related complications. [2-4] Hence, they may need to manage multiple prescribed medicines, fixed-dose combination products, over-the-counter medicines, nutritional supplements, and complementary health products as part of their daily care independently at home even if physicians check and modify treatments periodically. [2-4] Polypharmacy and treatment complexity are associated with medication non-adherence, drug interactions, adverse drug events, poor clinical outcomes, and increased healthcare expenditure. [2,3] There is a need for safer and more coordinated outpatient medication management that connects patients' daily experiences with the clinical consultation. Many existing applications function as reminder tools or static medication logs. [5] Also, when patients experience mild or moderate symptoms such as gastrointestinal discomfort or muscle pain, they may be uncertain whether a medicine is responsible. Fear of side effects may then contribute to unguided medication discontinuation. Patient information is stored on the patient's phone and is not available in a form that can be reviewed easily during a typical outpatient consultation. This creates a clear workflow and market gap, *i.e.*, patients lack an integrated tool for understanding medicines, documenting possible side effects, and preparing for consultations, and clinicians spend limited consultation time reconstructing medication use and events reported from home.

Target User/customer:

The primary users are adults with T2DM managing complex medicines at home, while clinicians are the secondary users who review the consultation summary. The likely institutional customers are hospitals, and diabetes clinics seeking to improve medication safety and consultation efficiency.

Product Idea:

CarePulse: Diabetes Safety & Doctor Consultation Companion is an AI-assisted outpatient companion with 3 integrated modules.

Module 1: *The Schedule Builder and Plain-Language Guide* converts supported prescriptions and fixed-dose combination products into a visual daily medication schedule. It also explains the active ingredients and intended purpose of each medicine in clear, patient-accessible language.

Module 2: *The Symptom Tracker and Summarizer* enables patients to record symptoms, their severity, onset, and relevant details. The system cross-references these entries with validated medication safety information to present possible drug *vs* symptom information and provide clear points for discussion with the clinician. It does not establish causality or recommend stopping treatment.

Module 3: *The Pre-Consultation Safety Snapshot* consolidates medication adherence, missed doses, newly added supplements, recorded symptoms, potential safety concerns, and patient questions into a concise summary that can be reviewed by the clinician before or during the consultation.

Why AI?

Artificial intelligence is appropriate for this use case because the underlying information is complex, unstructured, and context dependent. Prescription labels and drug monographs contain technical language that must be converted into understandable guidance. Regional brand names and multi-ingredient formulations must be mapped to their active generic ingredients. Patient reported symptoms are commonly expressed in free text and may require normalization before they can be compared with known adverse effects. Several medicines could be associated with a symptom. Hence, a rule-based approach becomes difficult to scale because it cannot readily interpret varied patient language, incomplete descriptions, temporal context, and multiple competing explanations. AI is therefore useful for language-based tasks such as symptom normalization, plain-language explanation, prioritization, and summarization, while validated data and deterministic rules remain the safety foundation. CarePulse works on a hybrid architecture in which validated medication data and deterministic safety rules provide the foundation, and language models support explanation, symptom organization, prioritization, and summarization. Since the product deals with medications, it incorporates a strict human-in-the-loop design. For the patient, CarePulse functions only as an educational and communication companion. Patients cannot start, stop, substitute, or change the dose of any medicine without direct clinical advice. For the clinician, the pre-consultation snapshot provides structured information but does not modify treatment plans or make diagnostic conclusions. All decisions concerning diagnosis, medication changes, and further clinical management remain exclusively under clinician authority.

Recommendation: Proceed to a controlled prototype. A limited pilot should be considered only after the prototype meets predefined medication-mapping, unsafe-action prevention, traceability, usability, privacy, and regulatory-readiness criteria.

Recommendation Summary:

Decision area	Recommendation	Rationale
Product direction	Proceed to a controlled prototype	The problem is recurring, the target users are identifiable, and the proposed workflow can be tested within a narrow scope.

Decision area	Recommendation	Rationale
Technical approach	Hybrid rules, validated retrieval, and language model	Rules and verified data protect medication safety. AI supports language interpretation, organization, and summarization.
Clinical autonomy	No autonomous clinical decision-making	Diagnosis, causality assessment, and treatment decisions remain with the clinician.
Pilot decision	Do not pilot yet	The prototype must first demonstrate medication-mapping accuracy, prevention of unsafe advice, summary traceability, and usability.
Scale decision	Defer	Scale should be considered only after a controlled pilot establishes safety, usefulness, privacy, and regulatory readiness.

Product Strategy and Opportunity Framing:

Product Vision:

To support adults with Type 2 Diabetes Mellitus managing complex medication regimens by improving understanding of prescribed therapies, facilitating safe documentation of medication-related concerns, and enabling more informed and efficient clinician-patient consultations.

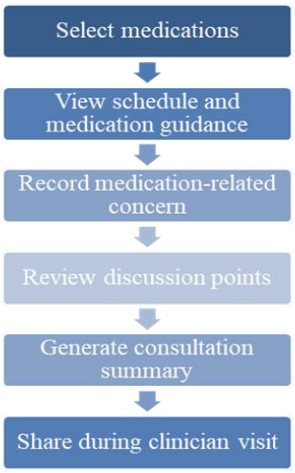
Jobs to be Done (JTBD):

Format used is When I ---- (context), who ---- (user), and outcome (functional job) -------

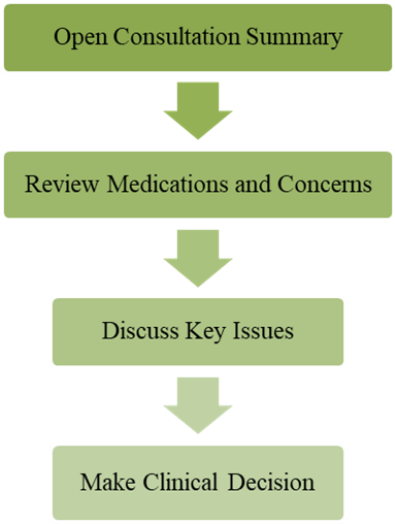
Persona	JTBD statement
Patient	When *I am managing multiple medicines at home*, I want *a clear understanding of my medicines and a simple way to keep track of any problems or side effects I notice*, so that *I can follow my treatment correctly and discuss concerns with my doctor*.
Clinician	When I am *reviewing a patient during a follow-up visit*, I want *a concise summary of medication use, patient-reported medication-related concerns,* so that *I can make informed clinical decisions easily.*

User Journey:

Patient Journey



Clinician Journey



Opportunity Framing:

Why does this problem matter? Managing Type 2 Diabetes commonly involves multiple medications, supplements, and treatment instructions that must be followed at home. Patients may find it difficult to understand their medication plan, recognize possible medication-related problems, and communicate relevant information during consultations. For clinicians, incomplete or fragmented information can make follow-up visits less efficient and increase the time spent reviewing medication histories and patient concerns.

Why are current alternatives insufficient? Most existing medication-management applications focus on reminders and medication tracking. They provide limited support for understanding complex medication regimens, documenting possible medication-related concerns, or preparing

information for discussion with clinicians. Hence, important information remains fragmented between the home environment and the clinical consultation.

Why might AI help? The workflow generates large amounts of information, including medication lists, adherence records, patient-reported concerns, and consultation information. AI can help organize this information by translating complex medication information into simpler language, providing a structured report of patient experiences, and generating concise summaries for clinical review. This improves communication and decision-making.

Key Assumptions:

1. Patients have access to a smartphone or web-enabled device.
2. Patients can identify the medications they are currently taking.
3. Clinicians are willing to review a short pre-consultation summary.

- Patients may write in English, in an Indian language, or in a mixture of both, including transliterated text.

Language Support: The MVP accepts free-text input in English, Hindi, and Telugu, including mixed-language and transliterated text. Patient-facing safety messages and medication guidance are reviewed and pre-approved before release rather than generated dynamically. This approach supports multilingual use and maintains consistency and safety for patient communication.

AI Opportunity Matrix:

Capability	Why?	Value	Feasibility	Priority
Medication schedule generation	Addresses the everyday challenge of managing multiple medicines and improves treatment understanding.	High	High	MVP
Plain language medication explanations	Help patients understand what each medicine is for and how it should be taken.	High	High	MVP
Brand-to-generic medication mapping	Reduces confusion caused by different brand names and combination medications.	High	Medium	MVP
Patient concern and side-effect tracking	Provides a structured way for patients to document issues experienced at home.	High	High	MVP
Medication related discussion points	Encourages informed patient-clinician discussions without making treatment decisions.	High	Medium	MVP

Capability	Why?	Value	Feasibility	Priority
Pre-consultation safety snapshot	Summarizes key information for clinicians and improves consultation efficiency.	High	High	MVP
Long-term adherence and symptom trends	May provide additional insights.	Medium	Medium	Future
Advanced analytics and predictive insights	Adds complexity and requires large datasets.	Medium	Low	Future
Diagnosis, cause assessment, or treatment recommendations	Clinical decisions must remain under clinician authority and fall outside the intended use of the product.	Unacceptable	Not pursued	Excluded

Build *vs* Buy *vs* API Assessment:

Capability	Recommendation	Rationale
Medication schedule builder	Build	This is an important part of the CarePulse workflow and directly addresses the patient's need to understand complex medication regimens.
Brand-to-generic medication mapping	Build	Accurate medication mapping is critical for safety and must be tailored to local brands and fixed-dose combinations.
Pre-consultation safety snapshot	Build	This is a key differentiator of the product and creates value for both patients and clinicians.
Plain-language medication explanations	API	AI text simplification is a widely available capability that can be accessed through existing language-model providers.
Patient concern organization and summarization	API	Existing language models can efficiently structure patient-reported concerns and generate summaries.

Recommended Path:

A hybrid Build + API approach is recommended. We shall build the capabilities that create the product's unique value, including the medication workflow, medication mapping layer, and pre-consultation safety snapshot. These capabilities directly support the goal of helping patients manage medications and helping clinicians review relevant information efficiently. We can use external APIs for language-based tasks such as simplifying medication information, organizing patient concerns, and generating summaries. These are commodity AI capabilities that do not provide significant competitive advantage when built from scratch. This allows us to focus our development effort on the parts of the product that differentiate CarePulse while reducing the time, cost, and infrastructure required to develop advanced AI models independently.

Main Trade-offs:

Option	Advantages	Trade-offs
Build	Greater control, customization, and ownership	Higher effort and longer development time
API	Faster implementation and lower infrastructure requirements	Dependence on external AI providers
Buy	Fastest deployment	Limited customization and poor alignment with the proposed workflow

Final Recommendation: Build the medication workflow, medication knowledge layer, and pre-consultation safety snapshot. Use external AI APIs for language understanding and summarization.

Medication Reference Data Sources:

Brand-to-generic mapping was assessed against publicly available drug reference sources before being assigned to Build. The assessment is summarised below.

Source	Access	Coverage	Use in CarePulse
CDSCO approved product information	Public website; no public API	Indian approvals, including fixed-dose combinations	Primary authority for the validated catalogue. Entries are extracted and verified manually.
Indian Pharmacopoeia Commission (IPC)	Public documents; no public API	Monographs and PvPI safety communications	Reference for ingredient standards and India-specific safety advisories.
Jan Aushadhi product list	Public website; no public API	Generic product names and strengths	Cross-check on generic naming and commonly dispensed presentations.

Source	Access	Coverage	Use in CarePulse
RxNorm / RxNav (US NLM)	Free public REST API, no licence fee	Normalised ingredient and strength concepts; limited Indian brand coverage	Ingredient-level normalisation and a structural model for the catalogue schema. Not the source of truth for Indian brands.
openFDA	Free public REST API	US label text and adverse reaction sections	Supporting reference for molecule-level adverse reaction profiles. Not patient-facing.
WHO ATC / DDD index	Public; non-commercial use	Therapeutic class hierarchy	Class-level grouping for regimen patterns.
MedDRA	Licensed via ICH MSSO; subscription required	Regulatory adverse reaction terminology	Target vocabulary for the lay-term mapping layer. Licensing and version maintenance are recorded as cost drivers.

Conclusion: Publicly available sources exist for ingredient normalisation (RxNorm) and for molecule-level label reference (openFDA), and both are adopted. but no reliable source provides comprehensive Indian brand-level data and fixed-dose combination coverage. Therefore, medication mapping is treated as a Build capability. To maintain accuracy and safety, the MVP is limited to 15 medication regimen patterns selected from the literature review, and medicines outside the supported catalogue are excluded from AI-assisted analysis.

First Wedge:

CarePulse will initially focus on adults with Type 2 Diabetes Mellitus who manage multiple medications at home and attend regular outpatient follow-up consultations. The first version of the product will address three core needs:

1. Understanding the medication plan through a clear daily schedule and plain-language guidance.
2. Documenting medication-related concerns and possible side effects between consultations.
3. Preparing a concise summary for discussion during the next clinician visit.

To keep the MVP focused and clinically safe, the initial version will support 15 medication regimen patterns selected for MVP scope based on the literature review commonly used by adults with Type 2 Diabetes Mellitus and associated comorbidities such as hypertension, cardiovascular disease, and chronic kidney disease. The MVP will also include a limited number of commonly reported medication-related concerns. The goal is to test whether patients can better understand

their treatment plans, document relevant information more consistently, and arrive at consultations better prepared for discussion with their clinicians.

Product Management Artifacts:

Problem Statement:

Adults with Type 2 Diabetes often manage multiple medications for diabetes and its associated comorbidities. As treatment complexity increases, patients may struggle to understand their medication plan, track medication-related concerns, and communicate relevant information during consultations. Existing solutions focus largely on reminders and tracking, leaving important information fragmented between home-based care and the clinician visit.

Product Vision:

To help individuals with Type 2 Diabetes Mellitus safely manage complex medication regimens, document medication-related concerns, and participate in consultations supported by a shared, structured understanding of their treatment experience.

Product Requirements Document:

Field	Description
User Problem	Adults with Type 2 Diabetes Mellitus and associated comorbidities managing multiple medications at home may find it difficult to understand medication plans, keep track of medication-related concerns, and communicate relevant information during clinical consultations.
Core Workflow	Understand medications -> Document concerns experienced between visits -> Share a structured summary during follow-up consultations.
Primary User	Adult with Type 2 Diabetes Mellitus managing multiple medications.
Secondary User	Diabetologist or internal medicine physician.
Objectives	O1. Improve medication understanding. O2. Enable structured, safe documentation of medication-related concerns. O3. Make consultations better prepared and faster. O4. Never generate unsafe medication advice.
Out of Scope	Diagnosis. Causality assessment. Dose or regimen changes. Treatment recommendations. EHR integration. Predictive analytics. Emergency monitoring. Any autonomous clinical decision.
Core Inputs	Medication list, medication schedule, supplement use, patient-reported concerns, adherence information, and consultation questions.

Field	Description
Core Outputs	Medication schedule, plain-language medication guidance, organized patient concerns, discussion points, and a pre-consultation summary.
AI Role	Simplify complex medication information, map patient free text onto a controlled symptom vocabulary, organize patient-reported concerns, generate discussion prompts, and create concise consultation summaries. The model selects from validated content; it does not originate clinical facts.
Success Measures	O1 is measured by Medication Schedule Understanding (>=80%). O2 by Summary Accuracy and Completeness (>=95%) and Unmapped Medication Rate. O3 by Consultation Preparedness (>=80%), Summary Usage Rate (>=80%) and Medication Reconciliation Time. O4 by Unsafe Medication Advice Rate (0%), Red-Flag Escalation Accuracy (100%) and Guardrail Activation Rate. Definitions and thresholds appear in the Evaluation Framework and Launch Thresholds sections.

Product Requirements Document - Risk, Safety and Fallback:

Field	Description
Key Risks	Incorrect medication mapping. Overstatement of medication-related concerns. Patient misunderstanding of AI outputs. Inappropriate reliance on the tool for clinical decision-making. Hallucinated or unsupported content in the pre-consultation snapshot. Data protection and unauthorised disclosure of health information. Automation bias, where a clinician or patient accepts an output without independent judgement. Scope creep into treatment advice.
Safety Boundaries	The product does not diagnose, determine causality, prescribe treatment, adjust medication doses, or replace clinical judgment.
Fallback Logic	If medication information cannot be verified, confidence is low, or urgent concerns are identified, the system does not generate recommendations and instead directs the user to seek clinician review or urgent medical attention.

Functional Requirements:

ID	Requirement	Rule	Acceptance Criterion	Obj.
FR-01	Medicine selection	Patient searches by brand or generic name, with fuzzy matching for common misspellings,	100% of catalogue entries return the correct product on exact and common-misspelling	O1

ID	Requirement	Rule	Acceptance Criterion	Obj.
		and selects from the validated catalogue. Patient enters prescribed frequency and timing.	search. Correct name, ingredients and strength are displayed.	
FR-02	Daily schedule generation	Deterministic rules convert the selected regimen and prescribed timing into a day view. No model involvement.	All selected medications appear in the correct time slot for 100% of catalogue regimens.	O1
FR-03	Plain-language medication guidance	Retrieval from the validated catalogue, rendered in reviewed plain-language text. Model output is constrained to retrieved content.	Patient can state the purpose and timing of each medicine; >=80% correct in usability testing. 0% of guidance contains content absent from the source record.	O1
FR-04	Concern and side-effect capture	Free-text entry with structured onset, severity and duration. Original wording is always stored unaltered.	Entries are saved with all structured fields and the verbatim original text retrievable in 100% of cases.	O2
FR-05	Symptom normalisation	The model maps free text onto the controlled lay-to-MedDRA vocabulary only. No term outside the curated set can be returned.	Mapping precision and recall measured against a reviewer-coded gold set. 0% of returned terms fall outside the controlled vocabulary.	O2
FR-06	Discussion point generation	A discussion point is produced only where the normalised term appears in the validated effect profile of a catalogue medicine. Possible-not-proven language is enforced.	Uncertainty language in 100% of outputs. 0% assert causality. Clinician Concern Rate below 20%.	O2, O4
FR-07	Unmapped medicine handling	Any medicine outside the validated catalogue is recorded, flagged for verification, and excluded from all AI-supported analysis.	100% of unmapped entries are excluded from analysis and surfaced to the clinician as unverified.	O2, O4

ID	Requirement	Rule	Acceptance Criterion	Obj.
FR-08	Red-flag identification	A deterministic high-recall screen runs before the language step. On a match, routine processing is bypassed and fixed urgent-care copy is shown.	100% of predefined urgent patterns escalated, in every supported language. No medicine is named as a cause in an urgent output.	O4
FR-09	Output guardrail	Generated text is screened for start, stop, substitute, dose-change and diagnostic language. A match suppresses the output and substitutes validated static content.	Unsafe Medication Advice Rate of 0%. Every suppression event is logged and reviewable.	O4
FR-10	Pre-consultation summary	Template-first assembly from patient-entered data and catalogue records. Every line is traceable to a source record.	Summary includes medication use, adherence, concerns, supplements and questions. >=95% of patient-entered information correctly represented. 0% unsupported statements.	O3
FR-11	Clinician review and override	The clinician can mark each discussion point relevant or not relevant. The judgement is recorded against the point.	Override decisions captured for 100% of reviewed points and feed the Clinician Concern Rate.	O3, O4
FR-12	Offline and degraded operation	Schedule and stored records remain available when AI-supported features are unavailable.	Schedule and stored entries render correctly with the language service disabled.	O1, O2

Non-Functional Requirements:

1. *Privacy:* Collect only the information required for the workflow and ensure patient information is handled securely.
2. *Reliability:* The medication schedule and patient records should remain accessible even if AI-generated features are unavailable.
3. *Usability:* Information should be presented using clear, patient-friendly language and simple interface design that can be easily understood during routine use.
4. *Safety:* The system must not provide diagnostic conclusions, medication changes, or treatment recommendations. All clinical decisions remain under clinician authority.
5. *Transparency:* AI-generated discussion points and summaries should be presented as informational support and should clearly communicate uncertainty when appropriate.
6. *Traceability:* Key outputs, such as medication information and safety messages, should be linked to validated reference sources.

PRD Functional vs Non-functional Requirement Mapping:

Functional Requirement	**Supporting Non-Functional Requirement**
Medication schedule	Reliability, Traceability
Plain-language guidance	Usability, Transparency
Concern tracking	Privacy, Usability
Discussion prompts	Safety, Transparency
Red-flag identification	Safety
Consultation summary	Reliability, Privacy, Transparency

Feature Prioritization:

Feature	**User Value**	**Effort**	**Confidence**	**Priority**	**Importance**
Medication Schedule Builder	High	Low	High	MVP	Addresses the core patient need of understanding complex medication regimens.
Plain-Language Medication Guidance	High	Medium	High	MVP	Helps patients understand what their medicines are for and how to use them safely.

Feature	User Value	Effort	Confidence	Priority	Importance
Brand-to-Generic Medication Mapping	High	Medium	Medium	MVP	Reduces confusion caused by multiple brand names and combination medications.
Patient Concern and Side-Effect Tracking	High	Low	High	MVP	Enables patients to document issues experienced between consultations.
Medication-Related Discussion Prompts	High	Medium	Medium	MVP	Supports patient-clinician communication without providing diagnosis or treatment advice.
Pre-Consultation Safety Summary	High	Medium	High	MVP	Help clinicians quickly review medication use and patient concerns before consultation.

MVP Definition:

The Minimum Viable Product (MVP) is a patient-facing prototype that demonstrates the complete CarePulse workflow from medication management at home to preparation for a clinical consultation.

The MVP includes three integrated modules:

Module 1: Schedule Builder and Plain-Language Guide - Patients can select medications from a validated medication catalogue and receive a structured daily medication schedule. The system also provides simple explanations of the purpose, ingredients, and key safety information associated with each medication.

Module 2: Symptom Tracker and Summarizer - Patients can document medication-related concerns, side effects, and other observations experienced between consultations. The system organizes these inputs and presents possible medication-related discussion points using appropriate uncertainty without establishing causality or recommending treatment changes.

Module 3: Pre-Consultation Safety Snapshot - The system generates a concise summary of medication use, adherence, newly added supplements, recorded concerns, and patient questions. This summary is intended to support more efficient and informed clinician-patient consultations.

To maintain a focused and clinically safe scope, the MVP will support 15 medication regimen patterns selected for MVP scope (based on the literature review) commonly used in the management of Type 2 Diabetes Mellitus and its associated comorbidities, along with a limited set of commonly reported medication-related concerns.

CarePulse MVP Catalogue - 15 Medication Product/Regimen Patterns

ID	Regimen / product pattern	Entry type	Catalogue mapping note
CP-01	Sitagliptin + Metformin + Pioglitazone FDC	Specified FDC	Three named ingredients.
CP-02	Dapagliflozin + Metformin FDC	Specified FDC	Two named ingredients.
CP-03	Sitagliptin + Metformin FDC	Specified FDC	Overlaps with the metformin-containing option in CP-12.
CP-04	Teneligliptin + Pioglitazone + Metformin FDC	Specified FDC	Same named ingredients as CP-11; retain a shared ingredient mapping.
CP-05	Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-06	Glibenclamide + Metformin	Named combination	FDC versus separate medicines not specified.
CP-07	Dapagliflozin + Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-08	Sitagliptin + Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-09	Pioglitazone + Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-10	Teneligliptin + Metformin	Named combination	FDC versus separate medicines not specified.
CP-11	Teneligliptin + Pioglitazone + Metformin as an insulin-resistance-focused regimen	Regimen scenario	User-supplied scenario label; same named ingredients as CP-04.
CP-12	Sitagliptin (with or without metformin)	Alternative regimen pattern	Distinguish sitagliptin alone from the metformin-containing option; overlaps with CP-03.

ID	Regimen / product pattern	Entry type	Catalogue mapping note
CP-13	SGLT2 inhibitor therapy alone or with DPP-4 inhibitor	Class-level pattern	Exact ingredients and product presentations not specified.
CP-14	ARB-based antihypertensive therapy in diabetic hypertension, especially ARB + CCB	Class-level pattern	Exact ingredients not specified; overlaps with CP-15.
CP-15	Telmisartan-based antihypertensive therapy, including telmisartan + amlodipine	Ingredient-based therapy pattern	Separate exact product records within this pattern; overlaps with CP-14.

Catalogue status: These are the 15 entries, not 15 distinct verified products. Overlapping entries are retained intentionally. Brand names, strengths, dosage forms, release characteristics, and source-verification records were not supplied and are not inferred. FDC means fixed-dose combination. The catalogue records scope only; it does not establish clinical suitability, approved indications, regulatory approval, or completed validation.

The MVP is designed to test whether:

1. Patients better understand their medication plans through structured schedules and plain-language explanations.
2. Patients can document medication-related concerns in a more organized and clinically useful manner.
3. Clinicians can review patient information more efficiently through a structured pre-consultation summary.

MVP Success Criteria:

Area	Success Criteria
Medication Accuracy	All supported medications display the correct ingredients and strengths from the validated medication catalogue.
Safety	The system does not advise patients to start, stop, or change medications.
Summary Accuracy	Information included in the pre-consultation summary is based on patient-entered or validated medication data.
Usability	Most participating adults with Type 2 Diabetes Mellitus can complete medication setup, concern entry, and pre-consultation summary preparation without assistance.
Clinician Usefulness	Clinicians find the pre-consultation summary useful for reviewing patient information before or during follow-up visits.

Roadmap:

Phase	Scope	Outcome
Phase 0: Discovery	Define the target workflow, identify key user problems, select 15 medication regimen patterns (based on the literature review) and document safety assumptions.	Problem, users, workflow, and safety boundaries are clearly defined.
Phase 1: MVP Prototype	Develop the 3 modules outlined.	A functional prototype demonstrates the complete patient-to-clinician workflow.
Phase 2: Usability Evaluation	Conduct simulated patient and clinician testing to evaluate clarity, usability, and usefulness of the workflow.	Evidence that participating adults with Type 2 Diabetes Mellitus can understand medications, document concerns, and use the consultation summary effectively.
Phase 3: Pilot Preparation	Refine the prototype, expand validated content, and complete privacy, security, safety, and regulatory review.	Product is considered suitable for limited real-world testing.
Phase 4: Controlled Pilot	Deploy CarePulse in a supervised outpatient setting without treatment recommendations or autonomous clinical decision-making.	Safety, usability, and workflow objectives are achieved in routine clinical use.
Phase 5: Scale Decision	Evaluate expansion of medication coverage, languages, and additional capabilities based on pilot findings.	Decision to scale is supported by positive clinical, operational, and governance outcomes.

North Star Metric: Percentage of patients reporting improved preparedness for medication related discussions with their clinician after using CarePulse.

Success Metrics:

Type	Metrics	Measure
Product	Regimen Mapping Accuracy	Percentage of supported medication regimens correctly mapped to their constituent ingredients and strengths.
	Medication Schedule Understanding	Percentage of patients who correctly understand the purpose and intended use of their medications after using the schedule and guide.

Type	Metrics	Measure
	Summary Completeness	Percentage of medication use, concerns, supplements, and patient questions accurately represented in the pre-consultation summary.
Business & Workflow	Medication Reconciliation Time	Reduction in consultation time spent reconstructing medication history and patient concerns.
	Potential Unguided Medication Discontinuation	Reduction in simulated scenarios where patients indicate they would stop treatment without clinician input.
	Summary Usage Rate	Percentage of consultations in which clinicians review the pre-consultation summary.
Safety & Trust	Red-Flag Escalation Accuracy	Percentage of predefined high-risk concerns correctly routed to clinician review or urgent care guidance.
	Guardrail Activation Rate	Frequency with which unsafe outputs are suppressed because they contain diagnostic, dosing, or medication-change recommendations.
	Unmapped Medication Rate	Percentage of medication entries that cannot be matched to the validated catalogue and require fallback handling.
	Clinician Concern Rate	Percentage of discussion prompts or summaries identified by clinicians as misleading, irrelevant, or potentially concerning.
Unit Economics	Cost per Enrolled Patient-Month	Inference, catalogue maintenance, clinical review, support and compliance cost divided by enrolled patients in the period.
Unit Economics	Inference Cost per Active Patient-Month	API spend attributable to language-model calls per active patient. Tracked separately because it is the cost line that scales with usage.
Unit Economics	Clinician Minutes Saved per Consultation	Reduction in medication reconciliation time, valued at loaded clinician cost, as the main quantifiable benefit.

Type	Metrics	Measure
Unit Economics	Cost per Consultation Successfully Prepared	Cost per consultation in which the clinician opened and used the snapshot. Links spend to realised value rather than to registrations.
Unit Economics	Catalogue Maintenance Cost per Regimen Added	Pharmacist review effort per new regimen. Determines whether catalogue expansion is economically sustainable.
Equity and Access	Task Success Rate by Age Band	Proportion of participating patients completing setup, concern entry and snapshot generation, reported for 30-44, 45-59, 60-74 and 75+.
Equity and Access	Task Success Rate by Digital Familiarity	Same measure, split by a self-reported smartphone-use proxy collected at enrolment.
Equity and Access	Task Success Rate by Health Literacy Proxy	Same measure, split by a short self-reported comprehension proxy. Reported as a proxy, not a clinical literacy assessment.
Equity and Access	Task Success Rate by Input Language	Same measure by language of entry, including code-mixed input.
Equity and Access	Red-Flag Escalation Accuracy by Language	Escalation accuracy reported separately per supported language rather than pooled.

UX, Workflow, and Trust Design:

Patient and Clinician Workflow:

Sno	Step	User Action	CarePulse Action	Safety Consideration
1	Medication Setup	Patient enters or selects medications from the supported catalogue.	Creates a structured medication plan and schedule.	Only validated medications are supported.
2	Medication Review	Patient views the medication schedule and plain-language guidance.	Explains medication purpose and key safety information.	Information is educational and does not replace clinician advice.
3	Daily Tracking	Patient records medication use and any concerns experienced between visits.	Stores medication and concern information in a structured format.	No clinical conclusions are generated at this stage.

Sno	Step	User Action	CarePulse Action	Safety Consideration
4	Concern Review	Patient reviews possible medication-related discussion points.	Organizes information to support discussion with the clinician.	Concerns are presented as possibilities, not confirmed causes.
5	Consultation Preparation	Patient generates a pre-consultation safety summary.	Summarizes medication use, concerns, supplements, and questions.	Information remains linked to validated medication data.
6	Clinical Consultation	Clinician reviews the summary during the visit.	Provides a concise overview of relevant patient information.	Clinical decisions remain entirely under clinician authority.

Wireframes and Prototype Screens:

Three design artefacts are included, moving from annotated layout through low-fidelity flow to visual treatment. All data shown in the screens is synthetic. The screens are illustrative and non-interactive; the working behaviour is demonstrated by the functional prototype described in the next section.

A. Home screen, annotated

The home screen carries three entry points and one persistent boundary statement. The intended-use line is placed in the main content area rather than in a footer or a one-time dialog, so the patient sees it on every visit rather than only at onboarding. On first use, the medicines area shows an empty state prompting the patient to add medicines, while concern recording stays available so that a patient can record something before completing setup.

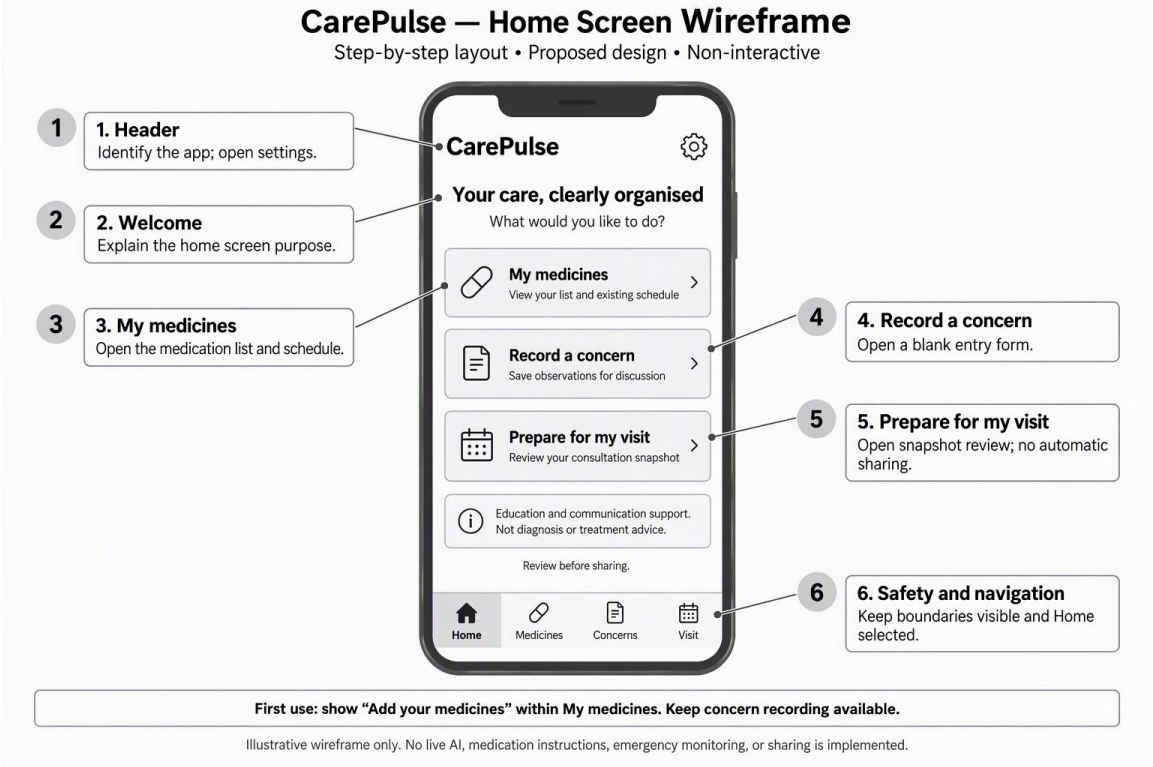


Figure 1. Home screen with numbered callouts. Illustrative wireframe. No live AI, medication instructions, emergency monitoring or sharing is implemented.

B. Low-fidelity screen flow

Six screens cover the full journey: medication setup, daily schedule, concern capture, discussion-point review, snapshot preparation and the clinician view. Two safety behaviours are visible in the layout itself. On screen 01, missing prescription instructions produce a verification prompt rather than an inferred default. On screen 04, an entry that cannot be interpreted returns an insufficient-information state in which the original entry is retained and no discussion point is generated.

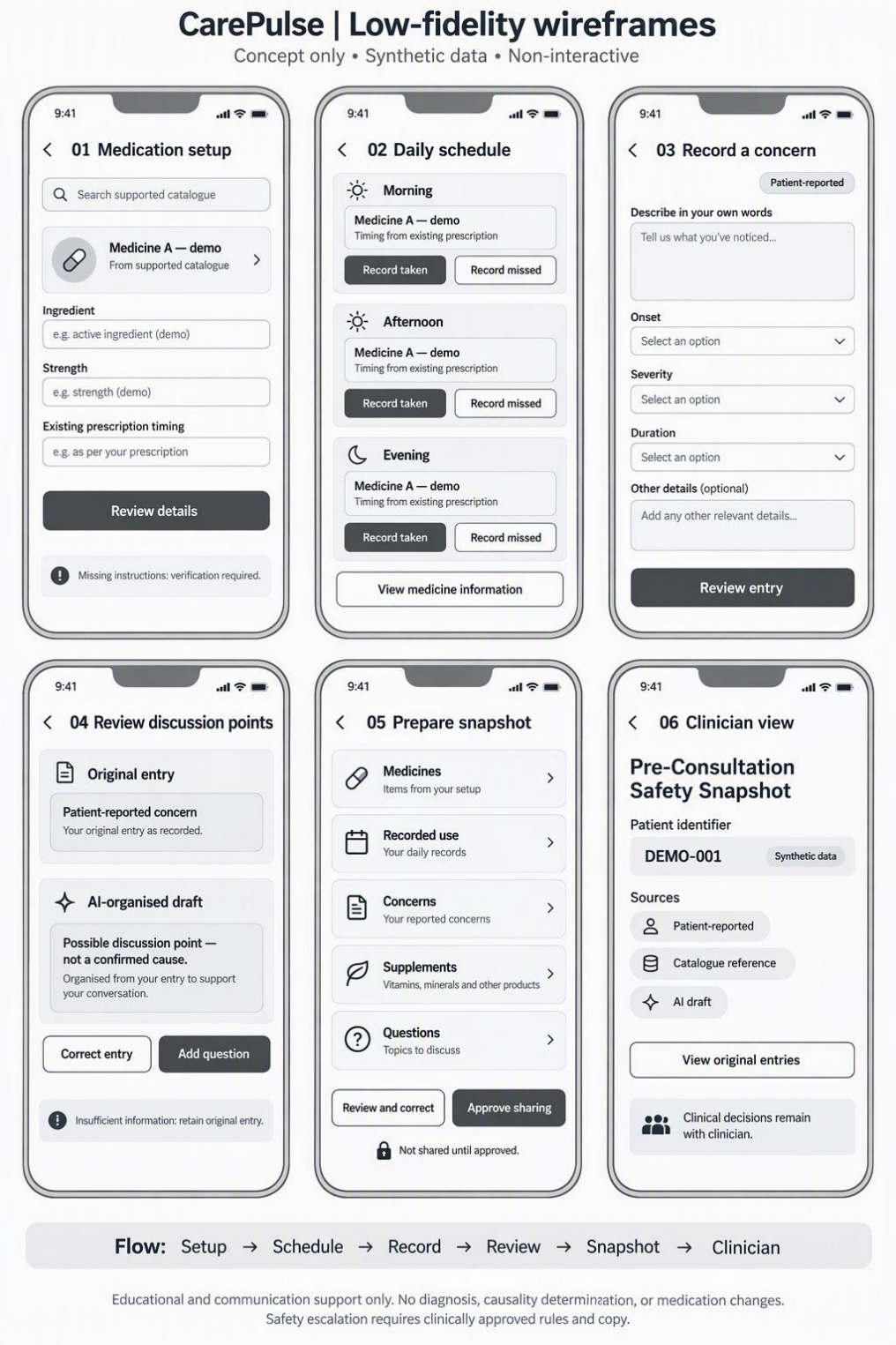


Figure 2. Low-fidelity wireframes. Flow: Setup -> Schedule -> Record -> Review -> Snapshot -> Clinician. Synthetic data, non-interactive.

C. Visual prototype screens

The visual treatment applies the uncertainty vocabulary consistently. Verified catalogue content, AI-organised drafts and missing information each carry a distinct and repeated visual marker, so a patient learns the distinction once and applies it everywhere. AI-drafted content is labelled as requiring review wherever it appears. Sharing is explicit: the snapshot is not released to the clinician until the patient approves it, and the locked state is shown on screen.

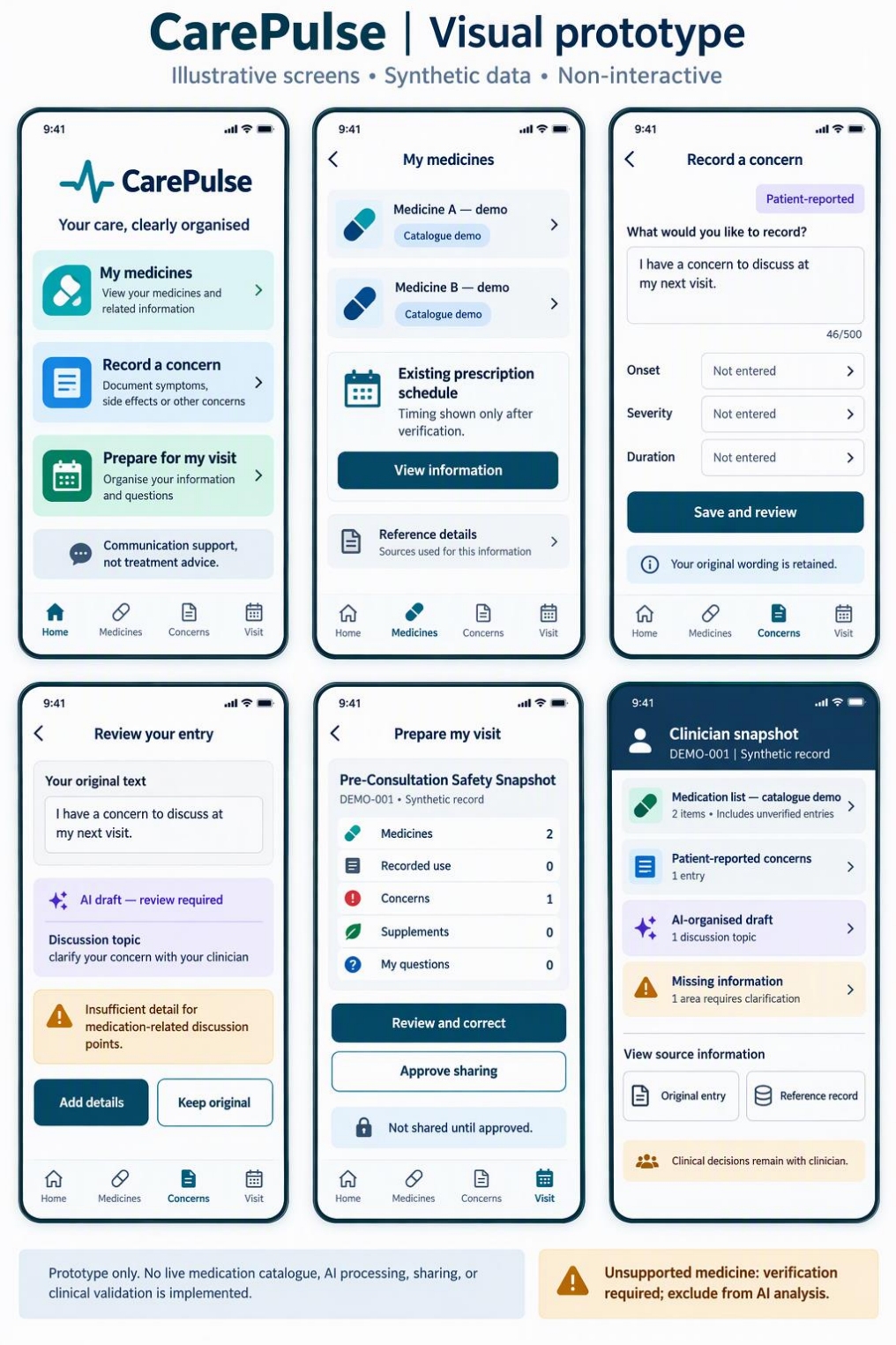


Figure 3. Visual prototype screens including the clinician snapshot view. Synthetic data, non-interactive.

Design decisions carried from the wireframes into the PRD

Screen behaviour	Requirement	Why it is designed this way
Intended-use statement on the home screen, not only at onboarding	Non-functional: Transparency	Reduces the risk that a returning patient treats the product as a diagnostic tool.
Missing prescription timing triggers a verification prompt	FR-01, FR-02	A schedule must never be produced from an inferred dose or timing.
Original patient wording always displayed alongside any AI draft	FR-04, FR-06	Keeps the patient able to detect a misreading, and preserves the verbatim record for the clinician.
Insufficient-information state with Add details and Keep original	FR-05, Fallback logic	The system withholds output rather than guessing, and the patient chooses whether to elaborate.
Unsupported medicine shown as unverified and excluded from analysis	FR-07	Prevents an unvalidated product from influencing any generated content.
Approve sharing step before the clinician can see the snapshot	Privacy control, Human-in-the-loop	Disclosure of health information stays under patient control.
Source links on the clinician view: original entry and reference record	FR-10, Traceability	Allows the clinician to check any line back to its source.
Relevant / Not relevant control on each discussion point	FR-11	Captures clinician judgement and generates the Clinician Concern Rate.

Functional Prototype / MVP

Field	Detail
Prototype type	Browser-based prototype, self-contained single HTML file. Language-model steps are simulated by a scripted rule engine (Wizard-of-Oz), declared on the prototype itself.
What it demonstrates	The complete workflow in both roles. Patient: daily schedule with dose recording, plain-language medicine guide, catalogue check for an unsupported medicine, concern entry, and the pre-consultation snapshot. Clinician: the snapshot, per-point relevant / not relevant review, and a live evaluation-metrics view.

Field	Detail
AI role made visible	Every generated output carries a safety trace naming the layer that produced or blocked it: validated catalogue (rule), red-flag screen (rule), language step (simulated), retrieval, and output guardrail. This distinguishes the product from a general chatbot.
Scenarios scripted	Path A, mapped and interpretable. Path B, medicine outside the catalogue, excluded and flagged. Path C, entry that cannot be interpreted, original wording retained. Path D, red flag, routine processing bypassed before the language step.
Catalogue in the prototype	Six curated regimens with ingredient mapping, timing, purpose, plain-language guidance and label-derived effect profiles. The full MVP scope is 15 regimen patterns as listed in the catalogue table.
What it does not prove	Real language-model behaviour on free text at scale. Mapping accuracy beyond the six loaded regimens. Lay-to-MedDRA mapping performance against a reviewer-coded gold set. That clinicians find the snapshot useful in a real clinic. Privacy, security or regulatory readiness. Cost per patient-month at volume.
Access	Single HTML file, opens in any browser with no account, server or network connection. Screenshots of the four scenario paths and the metrics view are included in the appendix.

The prototype is deliberately narrow. The open question for CarePulse is not whether a schedule can be rendered, but whether the product behaves safely when it is uncertain. The four scripted paths exist so that the failure behaviour can be inspected as directly as the success behaviour.

Trust and Explainability:

What does the patient need to know?

1. CarePulse is an educational and communication companion, not a diagnostic or treatment tool.
2. Discussion points are intended to support conversations with clinicians and should not be interpreted as confirmed medication effects.
3. Clinical decisions, medication changes, and treatment recommendations are always under clinician authority.

What must be visible?

1. The source of medication information and ingredient mapping.
2. Why a discussion point appears, such as medication information, symptom timing, recent changes, or repeated patterns.
3. Whether information is Verified Information, a Possible Discussion Point, or Insufficient Information.

4. Missing information, unsupported medications, and system limitations.
5. Clear guidance when clinician review or urgent medical attention is recommended.

What should be hidden?

1. Internal AI processes, model details, and technical calculations.
2. Confidence scores that may create a false impression of accuracy.
3. Information that could be interpreted as a diagnosis, treatment recommendation, or medication instruction.

Design Principles:

1. Always show the level of certainty and never hide uncertainty, exclusions, or missing information.
2. Present information in simple language that patients and clinicians can understand.
3. Provide clear correction options when medication information or patient-entered information appears inaccurate.
4. Keep urgent-care guidance visually distinct from routine discussion points.
5. When uncertainty exists, default to transparency and clinician review rather than generating potentially misleading conclusions.

Uncertainty and Fallback:

Sufficient Information Available: When medication information is available in the catalogue and sufficient patient information has been provided, CarePulse generates the medication schedule, plain-language guidance, discussion prompts, and pre-consultation summary. Information is presented as *verified information* and is linked to validated reference sources.

Information Uncertain or Incomplete: When information is incomplete, uncertain, or cannot be confidently interpreted, CarePulse requests additional details, retains the patient's original wording, and clearly communicates the limitation. Discussion points may be presented as *possible discussion points*, while unsupported conclusions and rankings are withheld. Examples include incomplete information about a medication-related concern, uncertain interpretation of patient-entered information or missing details required to generate meaningful discussion prompts.

Failure or Fallback Behaviour: When CarePulse cannot safely generate a response, it defaults to transparency and clinician review rather than making assumptions.

Situation	**Fallback Response**
Unmapped medication	Request verification and exclude the medication from AI-supported analysis.
Missing key information	Prompt the patient for additional details before continuing.
Uninterpretable patient concern	Retain the original entry and carry it forward to the consultation summary without AI-generated discussion points.

Situation	Fallback Response
Potentially urgent concern	Bypass routine processing and provide urgent-care or clinician-review guidance.
AI-generated content fails safety checks	Suppress the output and replace it with validated reference information or static guidance.
AI-supported features unavailable	Continue providing the medication schedule and stored patient information.

Design Principle: When uncertainty exists, CarePulse should explain what is known, what is unknown, and what action the user should take next, rather than presenting potentially misleading conclusions.

Human-in-the-Loop:

CarePulse is designed to support communication and decision-making, not to replace clinical judgment. Human oversight is built into the workflow to ensure patient safety and maintain clinician authority.

Stage	Human involvement
Medication Selection	Patients select medications from a validated catalogue and can review entries for accuracy before use.
Medication Understanding	Patients review medication information and decide what concerns or questions they wish to discuss with their clinician.
Concern Documentation	Patients record medication-related concerns using their own observations and experiences.
Discussion Preparation	CarePulse organizes information and presents possible discussion points, but patients decide which concerns are relevant to raise during consultation.
Consultation Review	Clinicians review the pre-consultation summary alongside their clinical assessment and patient history.
Clinical Decision-Making	All diagnosis, treatment decisions, medication changes, and follow-up plans remain entirely under clinician authority.

Data, Model, and Evaluation Strategy:

Problem as an AI Task:

CarePulse is not designed to diagnose disease or recommend treatment. Instead, it uses AI to support three tasks:

1. *Information Simplification:* Converting complex medication information into patient-friendly language.

2. *Information Organization:* Structuring patient-reported concerns into meaningful categories.
3. *Information Summarization:* Generating concise pre-consultation summaries for clinician review.

This solution combines information retrieval, language understanding, and controlled text generation.

Data Sources:

External Data Sources:

- CDSCO-approved medication information
- Indian Pharmacopoeia Commission (IPC) resources
- Jan Aushadhi Medicine Information
- Validated drug monographs and medication safety references

These sources provide medication names, ingredients, indications, and safety information.

User-Generated Data:

- Selected medications
- Medication schedule
- Adherence information
- Patient-reported concerns and side effects
- Supplement use
- Consultation questions

Internal Data:

- Validated medication catalogue covering the supported MVP regimens
- Medication-to-ingredient mappings
- Safety rules and escalation criteria

Data Assumptions:

The MVP assumes that patients can identify the medications they are currently taking, supported medications exist within the validated catalogue and patients can provide basic information about concerns they experience.

Labelling and Ground Truth:

The MVP does not require large-scale machine learning training. Ground truth will be derived from validated medication reference sources, expert-reviewed medication mappings, predefined symptom categories and known medication safety information. Patient-entered information will be used for workflow testing rather than model training.

Baselines:

CarePulse will be evaluated against both existing workflows and simpler non-AI alternatives.

Current Workflow Baseline: evaluates whether CarePulse improves the overall patient-clinician process.

Workflow	Current Approach	CarePulse Approach
Medication management	Reminder applications and medication lists	Structured schedule with plain-language guidance
Concern documentation	Manual notes or memory	Organized concern tracking
Consultation preparation	Patient recall during consultation	Pre-consultation safety summary

AI Capability Baseline: evaluates whether AI-assisted features provide additional value beyond simpler rule-based or manual alternatives.

AI Capability	Baseline Comparator
Plain-language medication guidance	Fixed clinician-written medication descriptions
Concern organization	User-selected predefined categories only
Discussion prompts	List of known side effects without prioritization
Pre-consultation summary	Chronological log export without summarization
Red-flag detection	Rule-based keyword matching

Model or API Choice:

A hybrid Rules + Retrieval + Language Model approach is recommended for CarePulse. We shall use validated medication data and rules for safety-critical decisions, retrieval for trusted medication information, and language models for explanation, organization, and summarization.

Rule-based approach:

1. Medication catalogue and ingredient mapping - Medication information must be accurate, traceable, and based on validated reference sources.
2. Medication schedule generation - Scheduling logic is structured and predictable and does not require AI reasoning.

Retrieval-based approach: Medication information retrieval - Ensures explanations are grounded in validated medication information rather than generated from memory

Language Model + Rule-based approach: Discussion point generation - Creates structured discussion prompts while maintaining safety guardrails and appropriate uncertainty.

Language Model + Retrieval-based approach: Plain-language medication guidance - Helps translate complex medical information into language that patients can easily understand.

Language Model-based approach:

1. Supports interpretation and organization of patient-entered concerns using controlled terminology mappings.
2. Pre-consultation safety summary - Summarizes patient information into a concise format for clinician review.

Evaluation Framework:

Technical Metric	**Threshold**	**Importance**
Medication Mapping Accuracy	100% for supported medications	Any mapping error can affect all downstream outputs.
Medication Schedule Understanding	>=80% of patients correctly identify medication purpose and timing	Measures whether the schedule and guidance improve understanding.
Summary Accuracy and Completeness	>=95% of patient-entered information correctly represented in the summary	Ensures clinicians receive reliable information.

Business/Workflow Metric	**Initial Threshold**	**Why It Matters**
Consultation Preparedness	>=80% of patients report feeling better prepared for consultation	Primary measure of product value.
Medication Reconciliation Time	Reduction compared with current workflow	Measures operational efficiency.
Summary Usage Rate	>=80% of clinicians review the summary during simulated consultations	Indicates adoption and relevance.
Clinician Usefulness Rating	>=4/5 from at least 80% of reviewers	Measures usefulness and signal-to-noise ratio.

Trust & Safety Metrics	**Initial Threshold**	**Why It Matters**
Unsafe Medication Advice Rate	0%	The system must never recommend starting, stopping, or changing medications.
Red-Flag Escalation Accuracy	100% for predefined urgent concerns	High-risk concerns must always be escalated appropriately.

Trust & Safety Metrics	Initial Threshold	Why It Matters
Unmapped Medication Rate	Monitored throughout testing	Indicates limitations of catalogue coverage.
Clinician Concern Rate	<20% of outputs flagged as misleading or irrelevant	Measures quality and relevance of generated outputs.
Guardrail Activation Rate	Monitored throughout testing	Confirms that safety controls remain active and effective.

Equity and Access Thresholds:

Equity / Economics Metric	Initial Threshold	Why It Matters
Task success by age band	No band more than 15 percentage points below the overall rate	The expected user base skews older. A product that works only for the youngest band has not met its own users.
Task success by digital familiarity	No group more than 15 percentage points below the overall rate	Low digital familiarity is a design problem to solve, not a user to exclude.
Task success by health literacy proxy	Monitored throughout testing; investigated if a gap exceeds 15 points	Reported as a proxy. Patients seek this product precisely because they are not clinically trained.
Red-flag escalation by language	100% per supported language, not pooled	A pooled figure can hide a language in which urgent expressions are missed.
Cost per enrolled patient-month	Measured and reported at pilot; no target set pre-pilot	Setting a target before measurement would be an unfounded claim.
Clinician minutes saved per consultation	Positive and measurable against the current-workflow baseline	If no time is saved, the institutional value proposition does not hold.

Subgroup reporting note: age bands and language are recorded directly. Digital familiarity and health literacy are short self-reported proxies collected at enrolment, not validated instruments, and are reported as such. A material gap triggers design investigation rather than exclusion of the affected group.

Test Design:

The MVP will be evaluated using a structured testing approach that combines functional, usability, and safety assessment. A predefined test set will include supported medication regimens, common

medication-related concerns, misspelled medication names, duplicate products, supplements, ambiguous patient-reported concerns, and urgent scenarios. Outputs will be reviewed by independent reviewers to assess accuracy, usefulness, and potential safety concerns. AI-assisted workflows will be compared with baseline approaches such as static medication information, manual concern tracking, and non-summarized consultation records. Evaluation will consider differences across users with varying levels of age, medication complexity, health literacy, and digital familiarity. Testing will be repeated whenever significant changes are made to the medication catalogue, safety rules, prompts, or AI components.

Evaluation Principles: CarePulse shall be assessed not only on technical performance, but also on whether it improves medication understanding, supports safer documentation of concerns, and helps patients and clinicians communicate more effectively during consultations.

Launch Thresholds:

The MVP will be considered ready to progress to the next phase only if it meets predefined requirements for accuracy, safety, usability, and clinician usefulness.

Area	Threshold	Go/No-Go Rationale
Medication Mapping Accuracy	100% accuracy for all supported medications in the catalogue	Any medication mapping error can affect all downstream outputs and is therefore unacceptable.
Unsafe Medication Advice	0% occurrence of recommendations to start, stop, substitute, or change medications	CarePulse must remain an educational and communication tool.
Pre-Consultation Summary Accuracy	>=95% of patient-entered information correctly represented in the summary	Clinicians must be able to trust the information presented.
Medication Schedule Understanding	>=80% of patients correctly identify medication purpose and timing	Demonstrates that the schedule and guidance improve understanding.
Consultation Preparedness	>=80% of patients report feeling better prepared for consultations	Primary validation of the product value proposition.
Clinician Usefulness Rating	>=4/5 rating from at least 80% of clinicians reviewing the summary	Indicates that the summary has sufficient signal-to-noise ratio for practical use.
Red-Flag Escalation Accuracy	100% of predefined urgent concerns correctly escalated	Any failure may have patient-safety implications and is a launch blocker.
Guardrail Performance	No unsafe output reaches the user without suppression or intervention	Confirms that safety controls are functioning as intended.

Area	Threshold	Go/No-Go Rationale
Lay-to-Clinical Term Mapping	Mapping quality meets the agreed precision and recall targets against the reviewer-coded gold set, in every supported language	Symptom normalisation is where patient language enters the safety logic. A failure here propagates to every downstream output.
Subgroup Task Success	No reported age or digital-familiarity group more than 15 percentage points below the overall task success rate	A product intended for older adults managing complex regimens must work for them, not only for the most digitally confident users.

Go/No-Go Decision:

CarePulse should proceed from MVP evaluation to pilot preparation only if:

- Medication information is consistently accurate.
- No unsafe medication advice is generated.
- Patients demonstrate improved medication understanding and consultation preparedness.
- Clinicians find the pre-consultation summary useful in reviewing patient information.
- Safety controls reliably identify and escalate urgent concerns.

Launch Principle:

CarePulse should move forward only when it demonstrates meaningful value for patients and clinicians while maintaining strict medication-safety boundaries. Safety and accuracy take precedence over feature completeness.

Business, Economics, and Scaling:

Monetization Model: Recommended monetization model is Enterprise/Clinic Subscription. The primary users of CarePulse are patients and clinicians, but the most likely paying customers are hospitals, diabetes clinics, and chronic disease management programs.

Customer	Value Provided	Monetization Approach
Hospitals and diabetes clinics	Improved consultation preparation and medication reconciliation	Annual or monthly institutional subscription
Chronic disease management programs	Better patient engagement and medication support	Per-enrolled patient or program-based subscription
Hospitals, Clinics	Scalable medication-management workflow	Enterprise licensing model
Individual patients	Access to medication support and consultation preparation tools	Free or low-cost patient access through participating healthcare organizations

Why This Model?

The primary value is created during the clinical encounter. Clinicians and healthcare organizations benefit from improved workflow efficiency. Patients are often reluctant to pay directly for medication-management tools. Institutional adoption supports broader and more sustainable deployment.

Main Trade-Offs

Option	Advantages	Limitations
Direct patient subscription	Simple to launch	May limit adoption and accessibility
Usage-based pricing	Matches activity levels	More complex to administer and predict
Enterprise subscription	Aligns with clinical workflow and organizational value	Requires institutional adoption and stakeholder buy-in
Hybrid model	Multiple revenue streams	Operationally more complex

Value Logic:

CarePulse creates value for patients, clinicians, and healthcare organizations by improving the flow of medication-related information between home and clinic.

- *Patient value:* Better understanding of medications, structured recording of concerns, and improved preparation for consultations.
- *Clinician value:* Less time spent reconstructing medication history and reviewing fragmented information during follow-up visits.
- *Healthcare Provider value:* More efficient consultations, improved continuity of care, and a differentiated chronic disease support service.
- *Organization value*: A reusable workflow that could potentially be extended to other chronic conditions involving complex medication management.

Value Proposition: CarePulse creates value by helping patients understand their medication plan, document concerns between visits, and arrive at consultations with better organized information for clinician review.

Cost Drivers:

Cost Area	MVP Approach	Scale Implication
Medication Catalogue Management	Manual validation of supported medication regimens	Additional medications require continued review and maintenance.

Cost Area	MVP Approach	Scale Implication
AI/API Usage	Limited use for explanation, organization, and summarization	Costs increase as the number of patients and AI supported interactions grows.
Clinical Review	Validation of medication information and safety rules	Ongoing governance and review processes may be required.
User Support	Self-service guidance and onboarding	Support requirements may increase with broader adoption.
Privacy and Security	Basic controls during MVP testing	Additional investment required before handling real patient data.

Unit Economics:

At the MVP stage, the objective is to demonstrate value rather than profitability. Long-term sustainability depends on whether the value created exceeds the operating cost of maintaining the platform.

Monthly value per enrolled patient = Clinician reconciliation minutes saved x loaded clinician cost + avoidable support contacts reduced x contact cost + retention/engagement value - monthly technology, review, support, and compliance cost. No financial savings claims should be made until they are demonstrated through pilot testing.

Unit Economics Principle: The workflow should create more value through improved consultation efficiency and patient support than it consumes in technology and operational costs.

Scaling Constraints and Risks:

As CarePulse expands beyond the MVP, several operational and technical challenges must be addressed.

- *Medication catalogue growth:* Additional medications, brands, and fixed-dose combinations will require ongoing validation and maintenance.
- *Regional medication variation:* Differences in local formulations and brand names increase medication-mapping complexity.
- *Language support:* Expanding into multiple languages requires careful review to preserve medical accuracy and readability.
- *Cost growth:* AI usage, medication catalogue maintenance, clinical review, user support, and compliance activities may increase operating costs as adoption grows.
- *Workflow complexity:* Supporting larger patient populations, more medication regimens, and additional use cases may increase product complexity and reduce usability if not carefully managed.
- *Governance burden:* Medication information, safety rules, AI outputs, and clinical risk controls require continuous monitoring, review, and updating.

- *Regulatory considerations:* Expanding the intended use toward diagnosis or treatment recommendations would substantially increase regulatory requirements and liability exposure.
- *System integration:* Future integration with electronic health records or hospital systems may create additional technical and operational challenges.
- *Latency and performance:* As usage grows, response times for AI-supported features and summary generation must remain acceptable for routine clinical use.
- *Alert fatigue:* Urgent escalation rules must maintain high sensitivity without generating excessive alerts that reduce clinician trust.

Scaling Principle:

CarePulse should scale only when medication accuracy, safety, usability, and governance can be maintained at the same standard achieved during the MVP and pilot stages. Growth should not compromise patient safety, clinician trust, or workflow simplicity.

Ethics, Governance, and Risk Note:

Main Risks:

1. *Unsafe or Misleading Outputs:* Patients may misunderstand discussion points as diagnosis, treatment advice, or confirmation of medication-related effects.
2. *Medication Information Errors*: Incorrect medication mapping or unsupported medications may lead to inaccurate guidance.
3. *Privacy and Data Protection:* Medication information, patient concerns, and consultation summaries contain sensitive health information that must be handled securely.
4. *Patient Over-Reliance*: Patients may treat the absence of a warning as reassurance and delay seeking medical advice.
5. *Clinician Adoption Risk*: If the consultation summary contains too much irrelevant information, clinicians may stop using it.

Harm Scenarios:

Potentially Affected Group	Possible Harm
Patients	Misinterpretation of discussion prompts as medical advice, delayed care, or inappropriate medication decisions.
Clinicians	Reliance on incomplete, inaccurate, or poorly summarized information during consultation.
Healthcare Organizations	Reputational, legal, or regulatory consequences if unsafe outputs reach users.

Risk Register:

Risk	Potential harm	Control	Launch blocker
Wrong ingredient mapping	Incorrect schedule or discussion	Validated catalogue; no free-form inference	Any unresolved supported-catalogue error
Overstated causality	Patient stops medicine	Possible-not-proven language; no action advice	Any unsafe instruction
Missed urgent concern	Delayed care	High-recall screen; urgent escalation copy	Sensitivity below threshold
Hallucinated snapshot	Clinician distracted or misled	Template-first summary; claim traceability	Unsupported statement rate above threshold
Privacy breach	Exposure of health information	Data minimization, encryption, access controls	Critical security issue
Bias / accessibility	Lower benefit or higher error for subgroups	Subgroup testing and language review	Material disparity without mitigation
Automation bias	Blind trust in output	Reason display, uncertainty labels, override capture	Clinician review workflow absent
Scope creep	Unvalidated clinical claims	Intended-use governance and change control	Product begins recommending treatment

Controls:

Bias and Fairness: Evaluate usability across different age groups, literacy levels, medication complexity, and levels of digital familiarity. Ensure that unsupported medications and missing information are handled consistently rather than inferred.

Safety Guardrails:

1. No diagnosis, treatment recommendations, or medication-change instructions.
2. Medication information limited to the validated catalogue.
3. Red-flag concerns are routed to clinician review or urgent-care guidance.
4. Unsupported medications are excluded from AI-supported analysis.

Privacy Controls:

1. Collect only the information required to support the workflow.
2. Separate patient identity from usage data wherever possible.
3. Apply appropriate security controls before any use of real patient information.

Review and Escalation:

1. Medication mappings and safety rules are validated before inclusion.
2. High-risk concerns bypass routine AI processing and trigger escalation guidance.
3. Clinicians remain responsible for all treatment decisions.

Logging and Traceability:

1. Medication information and summaries remain linked to validated reference sources.
2. System actions, safety-rule activations, and fallback decisions are traceable and reviewable.

Launch Blockers:

CarePulse should not proceed to pilot deployment if any of the following are identified:

1. Medication mapping errors within the supported catalogue.
2. Any output recommending medication initiation, discontinuation, substitution, or dose modification.
3. Failure to escalate predefined urgent concerns appropriately.

4. Significant inaccuracies in the pre-consultation summary.
5. Inability to clearly communicate uncertainty, limitations, or unsupported medications.
6. Privacy, security, or governance requirements not being met.

Governance Principle: CarePulse is intended to support medication understanding, communication, and consultation preparation. Patient safety, transparency, privacy, and clinician oversight take priority over feature expansion or deployment speed.

Role of Team Members:

Team Members	BITS ID	Role in Project Work
Ramyatha Munnangi	2025MB26036	Product Strategy and Research
Aishwarya M	2025MB26172	Clinical Workflow and Governance
Subhasree Dutta	2025MB26011	UX and Prototype
Anushka Tripathi	2025MB26168	PRD and Evaluation
Sumathi Senthil Kumar	2025MB26068	Business and Scaling

Each team member prepared and developed the sections related to their interest or domain following a team meeting to agree upon division of work. We had regular discussions, check-ins and comments wherein team members cleared doubts and suggested improvements. Drafts prepared by one team member were reviewed by others, and feedback was incorporated before finalizing the artifacts. Important decisions were discussed as a group and finalized after the entire team was ok with it.

AI Collaboration Log:

Tool: Claude (Anthropic)

Used for: Coding the functional prototype from the PM artifact, drafting the symptom-terminology design, researching free drug-reference APIs, compiling and verifying the reference list, document formatting.

Sample Prompt:

Build a functional browser-based prototype of CarePulse, the AI diabetes medication safety and consultation companion specified in our PM artifact. Use only what the document specifies. Do not invent clinical content.

SCOPE — build Module 2 and Module 3 end to end, plus the Module 1 schedule and plain language guide. Patient view and clinician view.

ARCHITECTURE — follow the hybrid stated in "Model or API Choice": Rules for the medication catalogue, ingredient mapping and schedule. Retrieval from validated catalogue records for

medication guidance. Language model only for symptom normalization, organizing concerns and summarizing. Simulate the language steps with a scripted engine so the demo runs without an API key, and declare that on the page.

CATALOGUE — six curated regimens as a subset of the 15 regimen patterns in our MVP catalogue table, each with brand, active ingredients and strengths, timing, purpose, plain language guidance and label-derived effect profile.

MUST DEMONSTRATE the four fallback behaviours from our "Uncertainty and Fallback" table, each as a scripted path: Unmapped medication: record, flag for verification, exclude from all AI-supported analysis, Missing key information: prompt for detail, Uninterpretable concern: retain original wording verbatim, generate nothing, carry forward to the clinician, Potentially urgent concern: bypass routine processing before the language, step, show fixed urgent-care guidance, name no medicine as a cause

SAFETY CHECKS — deterministic rules only, no model involvement: Same active ingredient appearing in more than one product, Daily ingredient total against the maximum stated in product information, Known interaction pairs from a curated table, including over-the-counter products and supplements. Where a logged symptom overlaps a flagged molecule, raise the entry to contact-your-doctor-soon.

ADVERSE EVENT ATTRIBUTION — where a normalized symptom appears in more than one medicine's effect profile, show every candidate with the date each was started. Do not rank, score or select a cause. Causality assessment is out of scope per our PRD.

GUARDRAILS from our Risk Register and Safety Guardrails: No output may recommend starting, stopping, substituting or changing a dose. Screen generated text for this language and suppress on a match. Possible-not-proven language on every discussion point. Uncertainty labels: Verified Information, Possible Discussion Point, Insufficient Information. Hide confidence scores that create false precision, per our "What should be hidden" list

TRACEABILITY — every output carries a visible trace naming which layer produced or blocked it: validated catalogue, red-flag screen, language step, retrieval, output guardrail.

CLINICIAN VIEW — the pre-consultation snapshot, a relevant / not relevant control on each discussion point to capture override, and a live metrics panel reading the thresholds in our Evaluation Framework.

CONSTRAINTS — single self-contained HTML file, no server, no account, synthetic data only, works offline. State on the page what the prototype does not prove.

References:

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Sources supporting the 15-regimen MVP catalogue

The catalogue scope was derived from an Embase literature search of Indian real-world studies, prescribing-pattern studies and guidelines, limited to 2000-2026 and to case reports, Cochrane reviews, controlled clinical trials, literature reviews, meta-analyses, observational studies, randomised controlled trials and systematic reviews. The search identified regimen patterns in use in India together with the medication-related concerns reported alongside them. The principal sources are listed below; the full search strategy and complete citation set are retained in the project record.

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